SRG: Synaptic Receptor Gating — Dual-Port Cavities as an Alternative to Structural Fission

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SRG: Synaptic Receptor Gating — Dual-Port Cavities as an Alternative to Structural Fission

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> *"A neuron does not split when it receives conflicting input. It receives it through different receptors."*

Abstract

When excitatory and inhibitory waves arrive at a single cavity in a wave-propagation substrate, they sum. Summation of opposed signals is destructive interference: the contradictory structure carried by the two waves cancels, and the plasticity updates they would have produced cancel with it. A concept holding genuine contradictions therefore learns nothing, because its updates annihilate.

Resonant Bifurcation solves this by splitting the cavity into two. This paper specifies the alternative: keep one cavity and **split its input ports**, exactly as a biological neuron uses distinct receptor types — AMPA for excitation, GABA for inhibition — on a single cell body.

Each cavity is given an **E-port** admitting only positive-polarity tunnels and an **I-port** admitting only negative-polarity ones. The two channels are integrated separately and combined only at readout, so signed cancellation cannot occur during accumulation. Graph topology stays fixed, no dynamic allocation is required, and the energy delivered by a query is not divided between child nodes.

The mechanism is implemented and opt-in, and §6 states honestly what that costs and when

the alternative is preferable.

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1. The Destructive Interference Problem

Consider a cavity receiving two tunnels, one of positive polarity and one of negative. In a scalar cavity, both contribute to the same displacement variable:

$$x \leftarrow x + \Delta_+ + \Delta_-$$

When $\Delta_+ \approx -\Delta_-$, the cavity is left at rest and the field concludes nothing happened. But something *did* happen: two structurally significant events arrived. The information was not processed and discarded — it was destroyed at the point of summation.

The consequence for learning is worse than the consequence for readout. Hebbian consolidation is driven by the product of endpoint velocities. If the velocity is cancelled, the product is near zero, and the tunnels that delivered the conflicting evidence are **not** strengthened. A concept that holds contradictions is therefore the one concept that cannot learn about them.

This is the phenomenon that motivated structural fission.

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2. Dual-Port Cavities

Rather than one intake, each cavity is given two:

- **E-port** — accepts waves from positive-polarity tunnels only.
- **I-port** — accepts waves from negative-polarity tunnels only.

The cavity therefore carries four state variables instead of two: (x_E, v_E) and (x_I, v_I).

The biological reading is direct. A cortical neuron does not maintain a single undifferentiated input. Excitatory and inhibitory afferents terminate on different receptor populations with different reversal potentials, and the cell integrates them through separate conductances before they jointly determine whether it fires.

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3. Mechanics

Waves entering the E-port and I-port do not sum into a shared displacement field during accumulation. They are integrated in separate channels, and the observable displacement is derived only at end-of-tick:

$$x = x_E - x_I, \qquad v = v_E - v_I$$

Equivalently, the I-port contribution is phase-inverted before it reaches the observable — a 180° shift — so an inhibitory signal carries an explicit negative functional sign rather than silently erasing an excitatory one.

The property that matters is where the subtraction happens. Both formulations produce the same observable x. They differ in what is available to plasticity: because Hebbian updates are computed from the *port* velocities rather than the observable, an excitatory update lands in v_E products and an inhibitory one in v_I products, and they cannot cancel. Structure accumulates in both channels simultaneously.

With gating disabled, x_I and v_I remain zero and the observable equals the E-port exactly, so the baseline dynamics are recovered bit-for-bit. The mechanism is a strict extension.

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4. Comparison with Structural Fission

Both mechanisms address the same failure. They differ in cost.

Property	Fission (RB+G)	Receptor gating (SRG)
Graph topology	mutates — new vertices spawned	**static**
Allocation	dynamic; parent/alias ledger required	none
Query energy	divided among children	**delivered whole to one cavity**
State per concept	grows	fixed, 2× scalar
Expressive gain	two independently-addressable concepts	one concept, two insulated pathways
Audit trail	explicit fission event with provenance	none — no structural event occurs

Static topology is the practical advantage. Fission requires allocating vertices at runtime and maintaining a ledger mapping a parent label to its children, so that later queries resolve correctly. Gating requires neither: the graph after a thousand ticks is the graph that was loaded.

Undivided energy is the functional advantage. Under fission, driving a concept excites a parent whose energy then distributes across children, so the amplitude reaching any single child is reduced. This was a live concern for recall gain. Under gating, the drive reaches one cavity at full amplitude and the tension is processed through two insulated pathways inside it.

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5. What Gating Does Not Provide

The comparison above would be dishonest without this section.

No new concepts. Fission produces two *addressable* entities: `metformin_alpha` can acquire its own edges, participate in its own analogies, and be reported separately. Gating produces one entity with better internal bookkeeping. If the contradiction is genuinely two distinct senses of a word, gating does not separate them — it only stops them destroying each other.

No structural audit trail. A fission event is a recorded structural mutation with provenance, and downstream analysis can ask *when* and *why* a concept split. Gating changes nothing structurally, so there is nothing to record and nothing to explain.

Doubled state. Every cavity pays for two ports whether or not it ever receives conflicting input. Fission pays only for concepts that actually contradict.

The mechanisms are therefore **complementary rather than competing**: gating is the cheap always-on defence against cancellation, fission is the expensive response to a contradiction severe enough to warrant two concepts.

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6. Implementation Status

Dual-port integration is implemented in the current substrate as the `x_E` / `v_E` and `x_I` / `v_I` channels on each cavity, with the observable derived at end-of-tick per §3.

It is **opt-in and off by default.** With SRG disabled the inhibitory channel stays at zero and the substrate reproduces baseline behaviour exactly, which is what makes the extension safe to carry. It is not enabled by default because the fission path is the mechanism for which reproducible multi-seed measurements exist; enabling gating by default would change the baseline against which those results were obtained.

An honest statement of the evidence: the cancellation problem of §1 is real and observed, the fix in §3 is sound by construction, and a head-to-head measurement of gating against fission on the same ontology **has not been run**. Until it is, the comparison table in §4 is an argument from mechanism, not a report of a result.

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7. Conclusion

Receptor gating is the conservative answer to destructive interference: it changes how a cavity integrates its inputs rather than how many cavities exist. It costs one extra pair of state variables per cavity, requires no dynamic allocation, and preserves the amplitude of a driven query.

It does not replace fission, because it cannot create a concept. It removes the reason a concept needed to be created in the *common* case — where the conflict is signed input to one coherent idea rather than two ideas sharing a name.

References

- Mahir, I. *RB+G: Gated Recursive Bifurcation with Analogical Bridging.* — the fission mechanism this paper offers an alternative to.
- Mahir, I. *From Artificial to Synthetic Intelligence.* §5.4c — the substrate carrying both mechanisms.
- Mahir, I. *Recursive Integrity Field Architecture (RIFA).* — the contradiction metabolism that detects the tension in the first place.
- Implementation: `src/physics.hpp` (`x_E`, `v_E`, `x_I`, `v_I`; end-of-tick observable derivation).